

# EMLM Group 11 Assignment

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Mixed effects model handles the data best, as the longitudinal model has correlated data.

What we are testing for: The main effect of Visit/time -> does UPDRS scores get worse over time

Age x Time interaction -> does age affect the rate of progression

Sex x Time interaction -> do men and women progress differently

Smoking status effect -> does smoking history matter?

Random effects variance -> how much do patients vary from each other?

```
library(nlme)
library(lattice)
library(reshape2)
library(ggplot2)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following object is masked from 'package:nlme':
##
## collapse

## The following objects are masked from 'package:stats':
##
## filter, lag

## The following objects are masked from 'package:base':
##
## intersect, setdiff, setequal, union

load("Group_11.RData")

head(DF)

##   id  UPDRS Visit  Age  sex smoking_status
## 1  1 17.1209     0 73.7 male              2
## 2  1 27.6828     1 73.7 male              2
## 4  1 69.1315     3 73.7 male              2
## 5  1 62.7264     4 73.7 male              2
## 6  1 68.4274     5 73.7 male              2
## 7  1 71.3014     6 73.7 male              2
```

```
str(DF)

## 'data.frame':    2762 obs. of  6 variables:
##  $ id           : Factor w/ 500 levels "1","2","3","4",...: 1 1 1 1 1 1 1
##  $ UPDRS        : num  17.1 27.7 69.1 62.7 68.4 ...
##  $ Visit        : num   0 1 3 4 5 6 7 0 3 6 ...
##  $ Age          : num  73.7 73.7 73.7 73.7 73.7 73.7 73.7 46.8 46.8 46.8
##  $ sex          : Factor w/ 2 levels "male","female": 1 1 1 1 1 1 1 2 2 2
##  $ smoking_status: int   2 2 2 2 2 2 2 0 0 0 ...

summary(DF)

##           id           UPDRS           Visit           Age           sex
##  1           :   7   Min.      : 0.22   Min.      :0.000   Min.      :41.10   male
##  6           :   7   1st Qu.: 13.98   1st Qu.:1.000   1st Qu.:52.60
##  11          :   7   Median : 22.73   Median :3.000   Median :65.00
##  12          :   7   Mean    : 26.02   Mean    :3.269   Mean    :64.63
##  14          :   7   3rd Qu.: 34.63   3rd Qu.:5.000   3rd Qu.:76.30
##  15          :   7   Max.     :106.94   Max.     :7.000   Max.     :88.00
##  (Other):2720
##  smoking_status
##  Min.      :0.0000
##  1st Qu.:0.0000
##  Median :0.0000
##  Mean    :0.6727
##  3rd Qu.:1.0000
##  Max.     :2.0000
##
##
##  # Fix sex and Smoking status as factors
DF$sex <- factor(DF$sex, levels = c('male', 'female'), labels = c('Male',
'Female'))

DF$smoking_status <- factor(DF$smoking_status, levels = c(0, 1, 2),
labels = c("Never", "Former", "Current"))
```

## Mean Structure

Question to answer: Does UPDRS tend to increase or decrease over time on average? Is the trend linear or non linear?

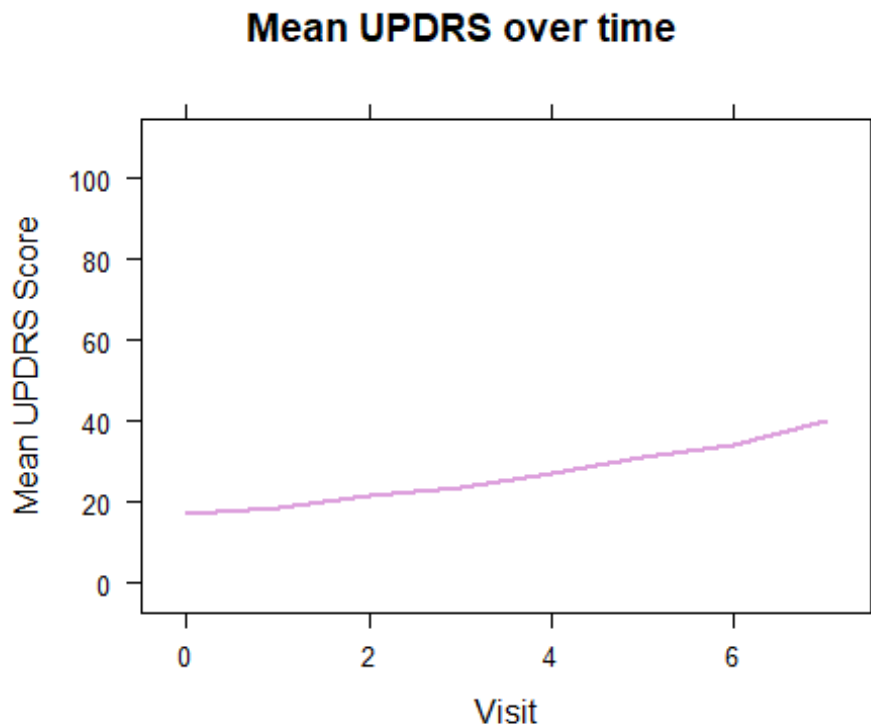
```
# Mean and SD of UPDRS at each visit
tapply(DF$UPDRS, DF$Visit, mean)

##           0           1           2           3           4           5           6           7
## 17.23940 18.83010 21.46871 23.47862 26.86689 31.06029 34.34414 39.98591
```

```
tapply(DF$UPDRS, DF$Visit, sd)

##           0           1           2           3           4           5           6
7
##  8.944131 10.888026 12.224334 13.400959 15.417547 18.321458 17.238031
18.591732

xyplot(UPDRS ~ Visit, data = DF,
       xlab = "Visit", ylab = "Mean UPDRS Score",
       main = "Mean UPDRS over time",
       panel = function(x, y) {
         panel.average(x, y, horizontal = FALSE, col = "plum", lwd = 2)
       })
```



The mean UPDRS score increases steadily across every visit, from 17.2 at visit 0 to 40 at visit 7. The increase looks roughly linear, with no obvious curve. Therefore, a linear time effect is a reasonable assumption.

## Variance Structure

Question to answer: Does the spread of UPDRS scores get larger or smaller over time?

```
# Variance by visit
var_by_visit <- tapply(DF$UPDRS, DF$Visit, var)

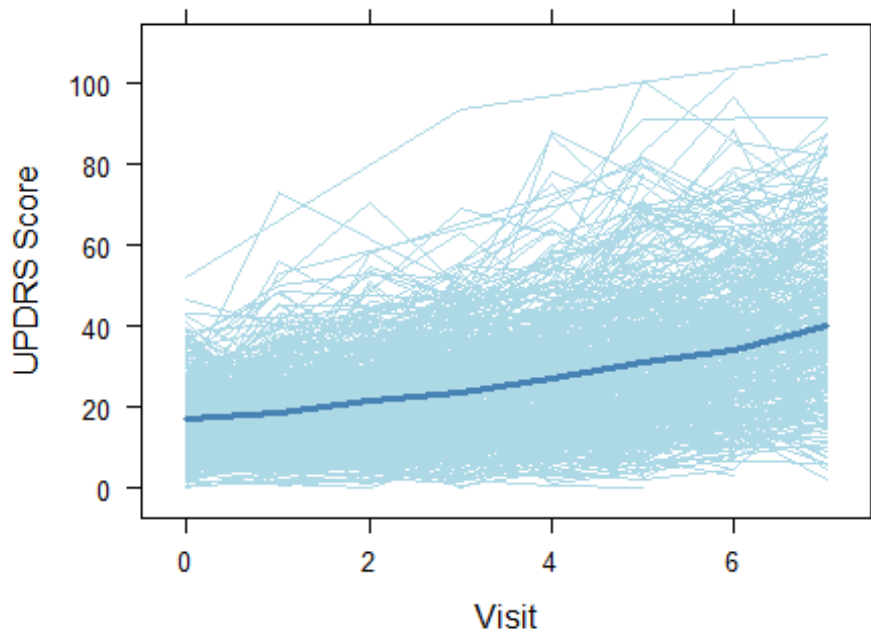
xyplot(UPDRS ~ Visit, groups = id, data = DF,
       type = "l", col = "lightblue",
```

```

xlab = "Visit", ylab = "UPDRS Score",
main = "Individual UPDRS profiles over time",
panel = function(x, y, ...) {
  panel.xyplot(x, y, ...)
  panel.average(x, y, horizontal = FALSE,
               col = "steelblue", lwd = 3)
})

```

### Individual UPDRS profiles over time

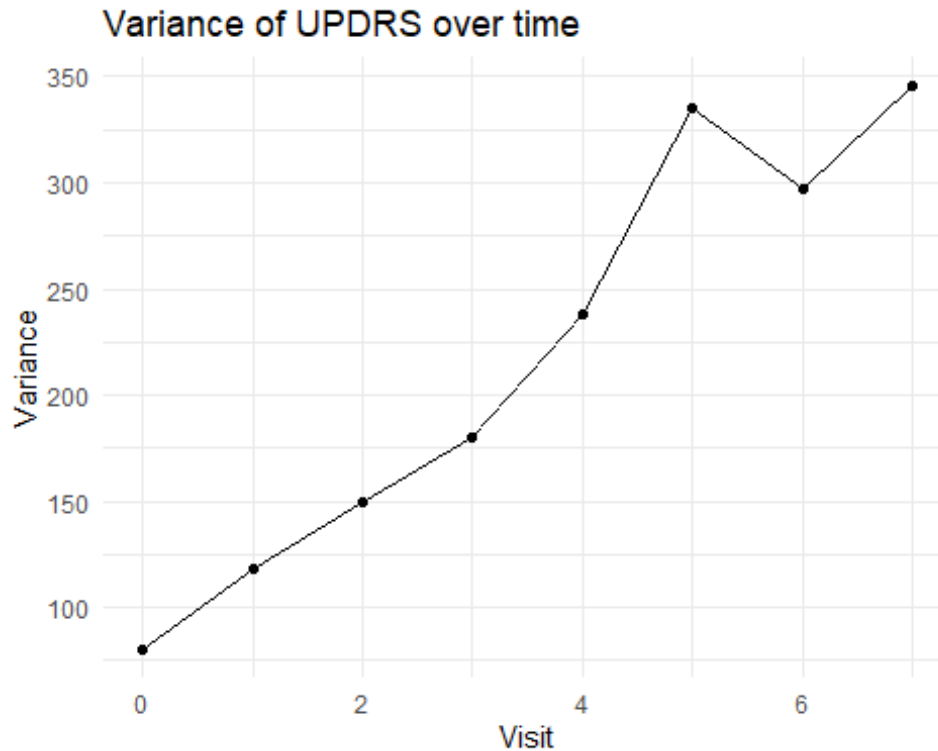


```

# Plot variance over time
visit_stats <- data.frame(
  Visit = 0:7,
  variance = as.numeric(var_by_visit),
  sd = as.numeric(tapply(DF$UPDRS, DF$Visit, sd))
)

ggplot(visit_stats, aes(x = Visit, y = variance)) +
  geom_line() +
  geom_point() +
  labs(title = "Variance of UPDRS over time",
       x = "Visit", y = "Variance") +
  theme_minimal()

```



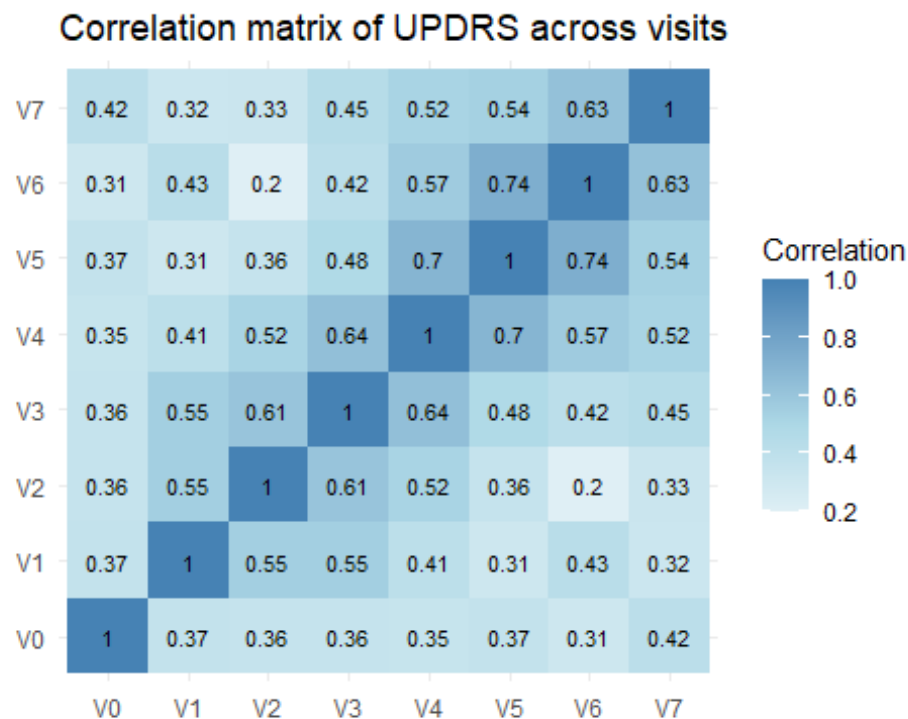
The variance increases from 80 at visit 0 and 346 at visit 7. This fanning effect shown in the spaghetti plot above tells us that the patients start at different levels at baseline, and they spread out over time. The progress rate is varied between patients. This tells us that both the baseline severity and the deterioration rates differ between patients. To capture this, we need to use both random slopes and random intercepts to model our data.

## Correlation Structure

Are measurements taken close together in time more similar to each other than measurements far apart?

```
# Wide format
DF_wide <- dcast(DF, id ~ Visit, value.var = "UPDRS")
cor_matrix <- cor(DF_wide[, -1], use = "pairwise.complete.obs")
colnames(cor_matrix) <- paste0("V", 0:7)
rownames(cor_matrix) <- paste0("V", 0:7)

# Heatmap
cor_melt <- melt(round(cor_matrix, 2))
ggplot(cor_melt, aes(x = Var1, y = Var2, fill = value)) +
  geom_tile() +
  geom_text(aes(label = value), size = 3) +
  scale_fill_gradient2(low = "white", high = "steelblue",
                      mid = "lightblue", midpoint = 0.5) +
  labs(title = "Correlation matrix of UPDRS across visits",
       x = "", y = "", fill = "Correlation") +
  theme_minimal()
```



The matrix above tells us that nearby visits correlate more strongly than distant ones. This decaying pattern rules out using compound symmetry, and suggests that a Toeplitz or AR(1) structure may work.

## Model Comparison

```
# Model 1: Random intercepts only
m_ri <- lme(UPDRS ~ Visit + Age + sex + smoking_status +
  Visit:smoking_status,
  random = ~1 | id,
  data = DF,
  method = "REML",
  na.action = na.exclude)

# Model 2: Random intercepts + slopes
m_ris <- lme(UPDRS ~ Visit + Age + sex + smoking_status +
  Visit:smoking_status,
  random = ~Visit | id,
  data = DF,
  method = "REML",
  na.action = na.exclude)

# LRT comparison
anova(m_ri, m_ris)
```

```
##           Model df          AIC          BIC      logLik    Test  L.Ratio p-value
## m_ri        1 10 21791.93 21851.14 -10885.97
## m_ris        2 12 21450.85 21521.90 -10713.42 1 vs 2 345.0881 <.0001
```

The LRT gives  $p < 0.0001$  and  $\delta AIC = 341$  in favour of the random slopes model. Note: We have used REML in this case, as we are comparing random effects structures.

## Compare Covariance Structures

```
m_ar1 <- gls(UPDRS ~ Visit + Age + sex + smoking_status +
Visit:smoking_status,
            data = DF,
            correlation = corAR1(form = ~Visit | id),
            method = "REML",
            na.action = na.exclude)
```

*# AIC/BIC comparison*

```
AIC(m_ri, m_ris, m_ar1)
```

```
##           df          AIC
## m_ri      10 21791.93
## m_ris     12 21450.85
## m_ar1     10 21598.19
```

```
BIC(m_ri, m_ris, m_ar1)
```

```
##           df          BIC
## m_ri      10 21851.14
## m_ris     12 21521.90
## m_ar1     10 21657.40
```

The above tells us that AR(1) performs better than random intercepts alone, but the random slopes model wins. Another advantage of choosing random slopes, is that the mixed effects approach with likelihood-based estimation is valid under MAR data. AR(1) does not have this advantage.

## Check Implied vs Observed Var-Cov

```
# random intercepts
getVarCov(m_ri, type = "marginal", individuals = 1)
```

```
## id 1
## Marginal variance covariance matrix
##           1          2          3          4          5          6          7
## 1 197.760   80.446   80.446   80.446   80.446   80.446   80.446
## 2  80.446  197.760   80.446   80.446   80.446   80.446   80.446
## 3  80.446   80.446  197.760   80.446   80.446   80.446   80.446
## 4  80.446   80.446   80.446  197.760   80.446   80.446   80.446
## 5  80.446   80.446   80.446   80.446  197.760   80.446   80.446
## 6  80.446   80.446   80.446   80.446   80.446  197.760   80.446
```

```
## 7  80.446  80.446  80.446  80.446  80.446  80.446 197.760
##   Standard Deviations: 14.063 14.063 14.063 14.063 14.063 14.063 14.063

# random slopes
getVarCov(m_ris, type = "marginal", individuals = 1)

## id 1
## Marginal variance covariance matrix
##      1      2      3      4      5      6      7
## 1 114.050  23.667  32.101  36.318  40.535  44.753  48.970
## 2  23.667 126.040  46.985  54.758  62.531  70.303  78.076
## 3  32.101  46.985 171.360  91.637 106.520 121.410 136.290
## 4  36.318  54.758  91.637 204.680 128.520 146.960 165.400
## 5  40.535  62.531 106.520 128.520 245.110 172.510 194.500
## 6  44.753  70.303 121.410 146.960 172.510 292.660 223.610
## 7  48.970  78.076 136.290 165.400 194.500 223.610 347.320
##   Standard Deviations: 10.68 11.227 13.09 14.307 15.656 17.107 18.636
```

From the above, the random intercepts model gives a flat matrix. This is clearly wrong, as our observed variances go from 80 to 346.

The random slopes model gives an increasing variance structure, which closely follows what we actually observed. This confirms that the random slopes model is correctly capturing the data's behaviour.